



System Documentation

Electronics

**Engine Management System ECS with Engine Interface Module
EIM 10**

- Engine Control Unit ECU 9 and Engine Monitoring Unit EMU 9
- Cylinder Pressure Monitor CPM
- Ignition Controller MIC 6
- Gas Regulating Unit GRU

**Application: Marine (gas engines)
8V4000M55RN, 16V4000M55RN**

E532473/02E



A Rolls-Royce
solution

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1 Safety

1.1 Important provisions for electronic products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Assembly, maintenance, and repair without ESD („Electro Static Discharge“) equipment and ESD work station
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

1.2 Intended use of electronic products

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original configuration of the delivery or in a configuration approved in writing by the manufacturer
- In compliance with all safety regulations and in adherence with all safety and warning notices in this manual
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:

- <http://www.mtu-solutions.com>

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

1.3 Intended use of gas engines in marine applications, engine with multi-point gas injection

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With preservation approved by the manufacturer in accordance with the (→ Preservation and Represervation Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (also applies to engine control/parameters)
- In compliance with all safety regulations and in adherence with all safety and warning notices in this manual
- In compliance with the maintenance work and intervals specified in the (→ Maintenance Schedule) throughout the useful life of the product
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair

The product must not be operated in explosive atmospheres unless the engine fulfills the conditions for such use and approval has been granted.

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:

- <http://www.mtu-solutions.com>

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

Gas system, general requirements

The engine must only be operated with the gas control path specified and delivered by Rolls-Royce Solutions. Any deviations from this requirement need to be checked separately. The gas control path must be arranged in the engine room in such a way that the connection from the gas control path to the engine, which is the scope of supply of the operator, complies with the specifications of Rolls-Royce Solutions (schematic drawing of fuel system and installation drawing of gas control path). Depending on load, the engine is operated at varying gaseous fuel pressure, which is regulated by the gas control path.

All components of the fuel system are designed for a maximum operating pressure of 10 bar. If pressure pulses cannot be excluded, the operator must install appropriate protective devices in the fuel supply to prevent pressure peaks.

The engine can be operated with two leak monitoring concepts:

- Nitrogen overpressure concept
- Air purge concept with at least 30 air changes per hour

When the nitrogen overpressure concept is applied, the fuel system is pressurized, with pressures up to 10 bar. Never loosen screws on the fuel system unless it was verified that the leak monitoring system had been brought to a safe (unpressurized) condition.

The gas control path can be operated exclusively with the air purge concept. Irrespective of the leak monitoring concept, the system integrator must supply the monitoring function as well as the media and sensors required for the leak monitoring function. When the engine is shut down, the Double Block and Bleed valve disconnects the gas supply to the engine. The gas supply pressure is still present in the gas control path then.